

Mühendislik Fakültesi

Bilgisayar Mühendisliği Bölümü

2019-2020 Bahar Yılı Uzakta Eğitim

Sınav Güncelleme Final Sınavı



2. Öğretim A subesisi



Soru 1)

(a) Aşağıda verilen tablonun son satırını $f'(x)$ için en iyi yaklaşımlar formüllerini kullanarak doldurunuz

* $f'(1,8)$ için 5 nokta orta nokta kullanılır. Sebabi ise hata oranı azdır

$$f'(1,8) = \frac{1}{12h} [-25f(x_0) + 48f(x_0+h) - 36f(x_0+2h) + 16f(x_0+3h) - 3f(x_0+4h)]$$

$$\frac{1}{12h} [-25f(1,8) + 48f(1,9) - 36f(2) + 16f(2,1) - 3f(2,2)] = \underline{3,833244}$$

* $f'(1,9)$ için 3 nokta orta nokta kullanılmalı. Anne 1,7 bilinmiyor.

$$f'(1,9) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)] \quad h=0,1$$

$$5[f(2,0) - f(1,8)] = \underline{2,953165}$$

* $f'(2,0)$ için 5 nokta orta nokta kullanılmalı

$$\frac{1}{12h} [f(x_0-2h) - 8f(x_0-h) + 8f(x_0+h) - f(x_0+2h)]$$

$$\frac{1}{12} [f(1,8) - 8f(1,9) + 8f(2,1) - f(2,2)] = \underline{2,249704}$$

* $f'(2,1)$ için 3 nokta orta nokta kullanılmalı

$$f'(x_0) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)]$$

$$5[f(2,2) - f(2,0)] = \underline{1,060958}$$

* $f'(2,2)$ için 5 nokta vs nokta kullanılmalı

$h = -0,1$ için

$$\frac{1}{12h} [-25f(x_0) + 48f(x_0+h) - 36f(x_0+2h) + 16f(x_0+3h) - 3f(x_0+4h)]$$

$$\frac{1}{12h} [-25f(2,2) + 48f(2,1) - 36f(2,0) + 16f(1,9) - 3f(1,1)]$$

$$= \underline{1,544209}$$

(b) $f(x) = xe^x$ fonksiyonu için $f''(2)$ türevini tabloyu kullanarak $h=0,1$ ve $h=0,2$ adımları ile yaklaşık hesapla

* $f'(2)$ ve $h=0,1$ için orta nokta

$$\frac{1}{h^2} [f(x_0-h) - 2f(x_0) + f(x_0+h)]$$

$$\frac{1}{(0,1)^2} [f(1,9) - 2f(2) + f(2,1)] = 29,5932$$

* $f''(2)$ ve $h=0,2$ için orta nokta

$$\frac{1}{h^2} [f(x_0-h) - 2f(x_0) + f(x_0+h)]$$

$$\frac{1}{(0,2)^2} [f(1,8) - 2f(2) + f(2,2)] = \underline{29,703525}$$

$$f'(2,0) = 1 \cdot e^x + e^x x = e^x(x+1)$$

$$f''(2,0) = e^x(x+2) = e^2(2+2) = \underline{29,5562}$$

* mutlak hata $h=0,1$ için

$$|29,5562 - 29,5932| = 3,6976 \times 10^{-2}$$

* Mutlak hata $h=0,2$ için

$$|29,5562 - 29,7035| = 1,49051 \times 10^{-1}$$

Soru 2 bir sonraki
sayfa da

Soru 2)

$$e^x + \frac{1}{2^x} + 2\cos x - 6 = 0$$

$$f'(x) = e^x - (12^{-x}) \cdot \ln 2 - 2\sin x$$

$$x_0 = 1$$

1. iterasyon

$$f(x_0) = f(1) = e^1 + \left(\frac{1}{2}\right) + 2\cos(1) - 6 = -1.70111356$$

$$f'(x_0) = f'(1) = e^1 - \left(\frac{1}{2}\right)(\ln 2) - 2\sin(1) = 0.68876627$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 1 - \frac{-1.70111356}{0.68876627}$$

$$\boxed{x_1 = 3.46979801}$$

2. iterasyon

$$f(x_1) = f(3.46979801) = e^{3.46979801} + \left(\frac{1}{2^{3.46979801}}\right) + 2\cos(3.46979801) - 6 \Rightarrow$$

$$= 24.3272653$$

$$f'(x_1) = f'(3.46979801) = e^{3.46979801} - \left(\frac{1}{2^{3.46979801}}\right)(\ln 2) - 2\sin(3.46979801) \Rightarrow$$

$$= 32.71237899$$

$$x_2 = 3.46979801 - \frac{24.3272653}{32.71237899}$$

$$\boxed{x_2 = 2.72612647}$$

3. iteration

$$f(x_2) = f(2.72612647) = e^{2.72612647} + \left(\frac{1}{2^{2.72612647}} \right) + 2\cos(2.72612647) - 6 \Rightarrow$$
$$= 7.59488418$$

$$f'(x_2) = f'(2.72612647) = e^{2.72612647} - \left(\frac{1}{2^{2.72612647}} \right) (\ln(2)) - 2\sin(2.72612647) \Rightarrow$$
$$= 14.36162032$$

$$x_3 = 2.72612647 - \frac{7.59488418}{14.36162032}$$

$$x_3 = 2.19729448$$

4. iteration

$$f(x_3) = f(2.19729448) = e^{2.19729448} + \left(\frac{1}{2^{2.19729448}} \right) + 2\cos(2.19729448) - 6 \Rightarrow$$
$$= 2.04605222$$

$$f'(x_3) = f'(2.19729448) = e^{2.19729448} - \left(\frac{1}{2^{2.19729448}} \right) (\ln(2)) - 2\sin(2.19729448) \Rightarrow$$
$$= 7.22931982$$

$$x_4 = 2.19729448 - \frac{2.04605222}{7.22931982}$$

$$x_4 = 1.91427308$$

5. iteration

$$f(x_4) = f(1.91427308) = e^{1.91427308} + \left(\frac{1}{2^{1.91427308}} \right) + 2 \cos(1.91427308) - 6 \Rightarrow$$
$$= 0.37378704$$

$$f'(x_4) = f'(1.91427308) = e^{1.91427308} - \left(\frac{1}{2^{1.91427308}} \right) (\ln(2)) - 2 \sin(1.91427308) \Rightarrow$$
$$= 4.71493221$$

$$x_5 = 1.91427308 - \frac{0.37378704}{4.71493221}$$

$$\boxed{x_5 = 1.8349958}$$

6. iteration

$$f(x_5) = f(1.8349958) = e^{1.8349958} + \left(\frac{1}{2^{1.8349958}} \right) + 2 \cos(1.8349958) - 6 \Rightarrow$$
$$= 0.02312695$$

$$f'(x_5) = f'(1.8349958) = e^{1.8349958} - \left(\frac{1}{2^{1.8349958}} \right) (\ln(2)) - 2 \sin(1.8349958) \Rightarrow$$
$$= 4.14022025$$

$$x_6 = 1.8349958 - \frac{0.02312695}{4.14022025}$$

$$\boxed{x_6 = 1.82940987}$$

7. iterasyon

$$f(x_6) = f(1.82940387) = e^{1.82940387} + \left(\frac{1}{2^{1.82940387}} \right) + 2\cos(1.82940387) - 6 \Rightarrow$$
$$= 0.00010776$$

$$f'(x_6) = f'(1.82940387) = e^{1.82940387} - \left(\frac{1}{2^{1.82940387}} \right) (\ln(2)) - 2\sin(1.82940387) \Rightarrow$$
$$= 4.10168047$$

$$x_7 = 1.82940387 - \frac{0.00010776}{4.10168047}$$

$$x_7 = 1.8293836$$

$$Hata = |x_7 - x_6| = 0.00002627156 < 10^{-4}$$

$$Kökümüz = x_7$$

$$x_7 \cong 1.82938360251$$

Soru 3)

Simpson yöntemi

$$\int_{0,8}^{1,2} \frac{\sin 2x}{x^2} dx \quad n=8 \quad h = \frac{b-a}{n} \Rightarrow h = \frac{1,2-0,8}{8} = 0,05$$

$$I \cong \int_a^b f(x) dx \cong \frac{0,005}{3} \left[F(0,8) + 4 [F(0,85) + F(0,95) + F(1,05) + F(1,15)] + 2 [F(0,9) + F(1) + F(1,1)] + F(1,2) \right]$$

$$F(0,8) = 1,5618337547523515$$

$$F(0,85) = 1,3725464504532434$$

$$F(0,9) = 1,2022810257755492$$

$$F(0,95) = 1,0485313531162482$$

$$F(1) = 0,909237426825681$$

$$F(1,05) = 0,782956341631631$$

$$F(1,1) = 0,6681988461318322$$

$$F(1,15) = 0,5638602738576327$$

$$F(1,2) = 0,4630716531605207$$

$$I \cong 0,3777000136$$

Soru 3)

Yamuk Yöntemi

$$\int_{0,8}^{1,2} \frac{\sin 2x}{x^2}$$

$$n=8$$

$$h = \frac{b-a}{n} \Rightarrow h = \frac{1,2-0,8}{8} = 0,05$$

$$I \cong \int f(x) dx \cong \frac{h}{2} [F(x_0) + 2[F(x_1) + F(x_2) + F(x_3) + F(x_4) + F(x_5) + F(x_6) + F(x_7)] + F(x_8)]$$

$$I \cong \frac{0,05}{2} [F(0,8) + 2[F(0,85) + F(0,9) + F(0,95) + F(1) + F(1,05) + F(1,1) + F(1,15)] + F(1,2)]$$

$$F(0,8) = 1,5618337547523515$$

$$F(0,85) = 1,3725464504532434$$

$$F(0,9) = 1,2022810257755492$$

$$F(0,95) = 1,0485319531162482$$

$$F(1) = 0,909297426825681$$

$$F(1,05) = 0,782956341631631$$

$$F(1,1) = 0,6681788461318922$$

$$F(1,15) = 0,5638602738576327$$

$$F(1,2) = 0,4690716531605207$$

$$I \cong 0,37815546319$$

Soru 3)

Dik Dörtgenler Yöntemi

$$\int_{0,8}^{1,2} \frac{\sin 2x}{x^2} dx$$

$$n=8$$

$$h = \frac{b-a}{n} = \frac{1,2 - 0,8}{8} = 0,05$$

$$I \approx \int_a^b f(x) dx \approx h [F(0,85) + F(0,9) + F(0,95) + F(1) + F(1,05) + F(1,1) + F(1,15)]$$

$$F(0,85) = 1,3725464504532439$$

$$F(0,9) = 1,2022810257755494$$

$$F(0,95) = 1,0485319531162487$$

$$F(1) = 0,9092974268256817$$

$$F(1,05) = 0,7829563416316314$$

$$F(1,1) = 0,6681788461318925$$

$$F(1,15) = 0,5638602738576332$$

$$I \approx 0,32738261588959405$$

Soru 4)

Euler metod

$$u_1 = u_0 + h f(x_0, u_0) = 1 + (0.025) f(0, 1) = 1 + (0.025) \cdot (0) \\ = 1$$

$$u_2 = u_1 + h f(x_1, u_1) = 1 + (0.025) f(0.025, 1) = 1 + (0.025) \cdot (0.03418) \\ = 1.00085$$

$$u_3 = u_2 + h f(x_2, u_2) = 1.00085 + (0.025) f(0.05, 1.00085) \\ = 1.00085 + (0.025) \cdot (0.06825) \\ = 1.00256$$

$$u_4 = u_3 + h f(x_3, u_3) = 1.00256 + (0.025) f(0.075, 1.00256) \\ = 1.00256 + (0.025) \cdot (0.1021) \\ = 1.00511$$

$$u_5 = u_4 + h f(x_4, u_4) = 1.00511 + (0.025) f(0.1, 1.00511) \\ = 1.00511 + (0.025) \cdot (0.13561) \\ = 1.0085$$

$$u_6 = u_5 + h f(x_5, u_5) = 1.0085 + (0.025) f(0.125, 1.0085) \\ = 1.0085 + (0.025) \cdot (0.16867) \\ = 1.01272$$

$$u_7 = u_6 + h f(x_6, u_6) = 1.01272 + (0.025) f(0.15, 1.01272) \\ = 1.01272 + (0.025) \cdot (0.20118) \\ = 1.01775$$

$$u_8 = u_7 + h f(x_7, u_7) = 1.01775 + (0.025) f(0.175, 1.01775) \\ = 1.01775 + (0.025) \cdot (0.23305) \\ = 1.02358$$

$$\begin{aligned} u_9 &= u_8 + h f(x_8, u_8) = 1.02358 + (0.025) f(0.2, 1.02358) \\ &= 1.02358 + (0.025) \cdot (0.26417) \\ &= 1.03018 \end{aligned}$$

$$\begin{aligned} u_{10} &= u_9 + h f(x_9, u_9) = 1.03018 + (0.025) f(0.225, 1.03018) \\ &= 1.03018 + (0.025) \cdot (0.29447) \\ &= 1.03754 \end{aligned}$$

$$\begin{aligned} u_{11} &= u_{10} + h f(x_{10}, u_{10}) = 1.03754 + (0.025) f(0.25, 1.03754) \\ &= 1.03754 + (0.025) \cdot (0.32388) \\ &= 1.04564 \end{aligned}$$

$$\begin{aligned} u_{12} &= u_{11} + h f(x_{11}, u_{11}) = 1.04564 + (0.025) f(0.275, 1.04564) \\ &= 1.04564 + (0.025) \cdot (0.35232) \\ &= 1.05445 \end{aligned}$$

$$\begin{aligned} u_{13} &= u_{12} + h f(x_{12}, u_{12}) = 1.05445 + (0.025) f(0.3, 1.05445) \\ &= 1.05445 + (0.025) \cdot (0.37874) \\ &= 1.06394 \end{aligned}$$

$$\begin{aligned} u_{14} &= u_{13} + h f(x_{13}, u_{13}) = 1.06394 + (0.025) f(0.325, 1.06394) \\ &= 1.06394 + (0.025) \cdot (0.40611) \\ &= 1.07409 \end{aligned}$$

$$\begin{aligned} u_{15} &= u_{14} + h f(x_{14}, u_{14}) = 1.07409 + (0.025) f(0.35, 1.07409) \\ &= 1.07409 + (0.025) \cdot (0.43137) \\ &= 1.08488 \end{aligned}$$

$$\begin{aligned} u_{16} &= u_{15} + h f(x_{15}, u_{15}) = 1.08488 + (0.025) f(0.375, 1.08488) \\ &= 1.08488 + (0.025) \cdot (0.4555) \\ &= 1.09626 \end{aligned}$$

$$\begin{aligned} u_{17} &= u_{16} + hf(x_{16}, u_{16}) = 1.09626 + (0.025)f(0.4, 1.09626) \\ &= 1.09626 + (0.025) \cdot (0.47847) \\ &= 1.10823 \end{aligned}$$

$$\begin{aligned} u_{18} &= u_{17} + hf(x_{17}, u_{17}) = 1.10823 + (0.025)f(0.425, 1.10823) \\ &= 1.10823 + (0.025) \cdot (0.50029) \\ &= 1.12073 \end{aligned}$$

$$\begin{aligned} u_{19} &= u_{18} + hf(x_{18}, u_{18}) = 1.12073 + (0.025)f(0.45, 1.12073) \\ &= 1.12073 + (0.025) \cdot (0.52094) \\ &= 1.13376 \end{aligned}$$

$$\begin{aligned} u_{20} &= u_{19} + hf(x_{19}, u_{19}) = 1.13376 + (0.025)f(0.475, 1.13376) \\ &= 1.13376 + (0.025) \cdot (0.54042) \\ &= 1.14727 \end{aligned}$$

$$\begin{aligned} u_{21} &= u_{20} + hf(x_{20}, u_{20}) = 1.14727 + (0.025)f(0.5, 1.14727) \\ &= 1.14727 + (0.025) \cdot (0.55875) \\ &= 1.16124 \end{aligned}$$

$$\begin{aligned} u_{22} &= u_{21} + hf(x_{21}, u_{21}) = 1.16124 + (0.025)f(0.525, 1.16124) \\ &= 1.16124 + (0.025) \cdot (0.57596) \\ &= 1.17564 \end{aligned}$$

$$\begin{aligned} u_{23} &= u_{22} + hf(x_{22}, u_{22}) = 1.17564 + (0.025)f(0.55, 1.17564) \\ &= 1.17564 + (0.025) \cdot (0.59201) \\ &= 1.19044 \end{aligned}$$

$$\begin{aligned} u_{24} &= u_{23} + hf(x_{23}, u_{23}) = 1.19044 + (0.025)f(0.575, 1.19044) \\ &= 1.19044 + (0.025) \cdot (0.60698) \\ &= 1.20561 \end{aligned}$$

$$\begin{aligned} u_{25} &= u_{24} + hf(x_{24}, u_{24}) = 1.20561 + (0.025)f(0.6, 1.20561) \\ &= 1.20561 + (0.025) \cdot (0.62088) \\ &= 1.22113 \end{aligned}$$

$$\begin{aligned} u_{26} &= u_{25} + hf(x_{25}, u_{25}) = 1.22113 + (0.025)f(0.625, 1.22113) \\ &= 1.22113 + (0.025) \cdot (0.63375) \\ &= 1.23658 \end{aligned}$$

$$\begin{aligned} u_{27} &= u_{26} + hf(x_{26}, u_{26}) = 1.23658 + (0.025)f(0.65, 1.23658) \\ &= 1.23658 + (0.025) \cdot (0.64561) \\ &= 1.25312 \end{aligned}$$

$$\begin{aligned} u_{28} &= u_{27} + hf(x_{27}, u_{27}) = 1.25312 + (0.025)f(0.675, 1.25312) \\ &= 1.25312 + (0.025) \cdot (0.65651) \\ &= 1.26953 \end{aligned}$$

$$\begin{aligned} u_{29} &= u_{28} + hf(x_{28}, u_{28}) = 1.26953 + (0.025)f(0.7, 1.26953) \\ &= 1.26953 + (0.025) \cdot (0.66647) \\ &= 1.28619 \end{aligned}$$

$$\begin{aligned} u_{30} &= u_{29} + hf(x_{29}, u_{29}) = 1.28619 + (0.025)f(0.725, 1.28619) \\ &= 1.28619 + (0.025) \cdot (0.67555) \\ &= 1.30308 \end{aligned}$$

$$\begin{aligned} u_{31} &= u_{30} + hf(x_{30}, u_{30}) = 1.30308 + (0.025)f(0.75, 1.30308) \\ &= 1.30308 + (0.025) \cdot (0.68377) \\ &= 1.32017 \end{aligned}$$

$$\begin{aligned} u_{32} &= u_{31} + hf(x_{31}, u_{31}) = 1.32017 + (0.025)f(0.775, 1.32017) \\ &= 1.32017 + (0.025) \cdot (0.69118) \\ &= 1.33745 \end{aligned}$$

$$\begin{aligned} u_{33} &= u_{32} + hf(x_{32}, u_{32}) = 1.33745 + (0.025)f(0.8, 1.33745) \\ &= 1.33745 + (0.025) \cdot (0.63782) \\ &= 1.3549 \end{aligned}$$

$$\begin{aligned} u_{34} &= u_{33} + hf(x_{33}, u_{33}) = 1.3549 + (0.025)f(0.825, 1.3549) \\ &= 1.3549 + (0.025) \cdot (0.70342) \\ &= 1.37249 \end{aligned}$$

$$\begin{aligned} u_{35} &= u_{34} + hf(x_{34}, u_{34}) = 1.37249 + (0.025)f(0.85, 1.37249) \\ &= 1.37249 + (0.025) \cdot (0.70832) \\ &= 1.39021 \end{aligned}$$

$$\begin{aligned} u_{36} &= u_{35} + hf(x_{35}, u_{35}) = 1.39021 + (0.025)f(0.875, 1.39021) \\ &= 1.39021 + (0.025) \cdot (0.71347) \\ &= 1.40805 \end{aligned}$$

$$\begin{aligned} u_{37} &= u_{36} + hf(x_{36}, u_{36}) = 1.40805 + (0.025)f(0.9, 1.40805) \\ &= 1.40805 + (0.025) \cdot (0.7174) \\ &= 1.42599 \end{aligned}$$

$$\begin{aligned} u_{38} &= u_{37} + hf(x_{37}, u_{37}) = 1.42599 + (0.025)f(0.925, 1.42599) \\ &= 1.42599 + (0.025) \cdot (0.72074) \\ &= 1.444 \end{aligned}$$

$$\begin{aligned} u_{39} &= u_{38} + hf(x_{38}, u_{38}) = 1.444 + (0.025)f(0.95, 1.444) \\ &= 1.444 + (0.025) \cdot (0.72352) \\ &= 1.46209 \end{aligned}$$

$$\begin{aligned} u_{40} &= u_{39} + hf(x_{39}, u_{39}) = 1.46209 + (0.025)f(0.975, 1.46209) \\ &= 1.46209 + (0.025) \cdot (0.7258) \\ &= 1.48024 \end{aligned}$$

$$u(1) = 1.48024$$

4. dereceden R-K metodu

$$k_1 = hf(x_0, u_0) = (0.1)f(0, 1) = (0.1) \cdot (0) = 0$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, u_0 + \frac{k_1}{2}\right) = (0.1)f(0.05, 1) = (0.1) \cdot (0.06827) \\ = 0.00683$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, u_0 + \frac{k_1}{2}\right) = (0.1)f(0.05, 1.100341) = (0.1) \cdot (0.13555) \\ = 0.00682$$

$$k_4 = hf(x_0 + h, u_0 + k_3) = (0.1)f(0.1, 1.00682) = (0.1) \cdot (0.13555) \\ = 0.01355$$

$$u_1 = u_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_1 = 1 + \frac{1}{6} [0 + 2(0.00682) + 2(0.00682) + (0.01355)]$$

$$u_1 = 1.00681$$

$$\underline{u(0.1) = 1.00681}$$

R-K metodunun
derecesi



$$k_1 = hf(x_1, u_1) = (0.1) f(0.1, 1.00681) = (0.1) \cdot (0.13555) \\ = 0.013555$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, u_1 + \frac{k_1}{2}\right) = (0.1) f(0.15, 1.01353) \\ = (0.1) \cdot (0.20114) \\ = 0.02011$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, u_1 + \frac{k_2}{2}\right) = (0.1) f(0.15, 1.01687) \\ = (0.1) \cdot (0.20036) \\ = 0.0201$$

$$k_4 = hf(x_1 + h, u_1 + k_3) = (0.1) f(0.2, 1.0269) \\ = (0.1) \cdot (0.26393) \\ = 0.02639$$

$$u_2 = u_1 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_2 = 1.00681 + \frac{1}{6} [0.01355 + 2(0.02011) + 2(0.0201) + (0.02639)]$$

$$u_2 = 1.02687$$

$$u(0.2) = 1.02687$$

R-K method
done



$$k_1 = hf(x_2, u_2) = (0.1)f(0.2, 1.02687) = (0.1) \cdot (0.26393) \\ = 0.02639$$

$$k_2 = hf\left(x_2 + \frac{h}{2}, u_2 + \frac{k_1}{2}\right) = (0.1)f(0.25, 1.04007) \\ = (0.1)(0.32365) \\ = 0.03237$$

$$k_3 = hf\left(x_2 + \frac{h}{2}, u_2 + \frac{k_2}{2}\right) = (0.1)f(0.25, 1.04305) \\ = (0.1) \cdot (0.32339) \\ = 0.03234$$

$$k_4 = hf(x_2 + h, u_2 + k_3) = (0.1)f(0.3, 1.05921) = (0.1) \cdot (0.37925) \\ = 0.03792$$

$$u_3 = u_2 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_3 = 1.02687 + \frac{1}{6} [0.02639 + 2(0.03237) + 2(0.03234) + (0.03792)]$$

$$u_3 = 1.05916$$

$$\underline{y(0.3) = 1.05916}$$

R-K method
done!
↓

$$k_1 = hf(x_3, u_3) = (0.1) f(0.3, 1.05916) = (0.1) \cdot (0.37925) \\ = 0.03793$$

$$k_2 = hf\left(x_3 + \frac{h}{2}, u_3 + \frac{k_1}{2}\right) = (0.1) f(0.35, 1.07812) = (0.1) \cdot (0.43083) \\ = 0.04303$$

$$k_3 = hf\left(x_3 + \frac{h}{2}, u_3 + \frac{k_2}{2}\right) = (0.1) f(0.35, 1.0807) = (0.1) \cdot (0.43058) \\ = 0.04306$$

$$k_4 = hf(x_3 + h, u_3 + k_3) = (0.1) f(0.4, 1.10222) = (0.1) \cdot (0.47768) \\ = 0.04777$$

$$u_4 = u_3 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_4 = 1.05916 + \frac{1}{6} [0.03793 + 2(0.04303) + 2(0.04306) + (0.4777)]$$

$$u_4 = 1.10215$$

$$\underline{u(0.4) = 1.10215}$$

$$k_1 = hf(x_4, u_4) = (0.1) f(0.4, 1.0215) = (0.1) \cdot (0.47769) \\ = 0.04777$$

$$k_2 = hf\left(x_4 + \frac{h}{2}, u_4 + \frac{k_1}{2}\right) = (0.1) f(0.45, 1.12604) = (0.1) \cdot (0.52016) \\ = 0.05202$$

$$k_3 = hf\left(x_4 + \frac{h}{2}, u_4 + \frac{k_2}{2}\right) = (0.1) f(0.45, 1.12816) = (0.1) \cdot (0.51985) \\ = 0.05199$$

$$k_4 = hf(x_4 + h, u_4 + k_3) = (0.1) f(0.5, 1.15414) = (0.1) \cdot (0.55766) \\ = 0.05577$$

$$u_5 = u_4 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_5 = 1.10215 + \frac{1}{6} [0.04777 + 2(0.05202) + 2(0.05199) + (0.05577)]$$

$$u_5 = 1.15408$$

$$\underline{y(0.5) = 1.15408}$$

$$k_1 = hf(x_5, u_5) = (0.1)f(0.5, 1.15408) = (0.1) \cdot (0.55767) \\ = 0.05577$$

$$k_2 = hf\left(x_5 + \frac{h}{2}, u_5 + \frac{k_1}{2}\right) = (0.1)f(0.55, 1.18196) = (0.1) \cdot (0.53094) \\ = 0.05303$$

$$k_3 = hf\left(x_5 + \frac{h}{2}, u_5 + \frac{k_2}{2}\right) = (0.1)f(0.55, 1.18362) = (0.1) \cdot (0.53066) \\ = 0.05307$$

$$k_4 = hf(x_5 + h, u_5 + k_3) = (0.1)f(0.6, 1.21314) = (0.1) \cdot (0.61953) \\ = 0.06195$$

$$u_6 = u_5 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_6 = 1.15408 + \frac{1}{6}[0.05577 + 2(0.05303) + 2(0.05307) + (0.06195)]$$

$$u_6 = 1.21308$$

$$u(0.6) = 1.21308$$

$$k_1 = hf(x_0, u_0) = (0.1)f(0.6, 1.21308) = (0.1) \cdot (0.61954) \\ = 0.06195$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, u_0 + \frac{k_1}{2}\right) = (0.1)f(0.65, 1.24406) = (0.1) \cdot (0.66428) \\ = 0.06643$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, u_0 + \frac{k_2}{2}\right) = (0.1)f(0.65, 1.2453) = (0.1) \cdot (0.66405) \\ = 0.06644$$

$$k_4 = hf(x_0 + h, u_0 + k_3) = (0.1)f(0.7, 1.27749) = (0.1) \cdot (0.66649) \\ = 0.06649$$

$$u_2 = u_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_2 = 1.21308 + \frac{1}{6}[0.06195 + 2(0.06643) + 2(0.06644) + 0.06649]$$

$$u_2 = 1.27744$$

$$u(0.7) = 1.27744$$

$$k_1 = hf(x_2, u_2) = (0.1) f(0.7, 1.27744) = (0.1) \cdot (0.66432) \\ = 0.06643$$

$$k_2 = hf\left(x_2 + \frac{h}{2}, u_2 + \frac{k_1}{2}\right) = (0.1) f(0.75, 1.31068) = (0.1) \cdot (0.68223) \\ = 0.06822$$

$$k_3 = hf\left(x_2 + \frac{h}{2}, u_2 + \frac{k_2}{2}\right) = (0.1) f(0.75, 1.31155) = (0.1) \cdot (0.68205) \\ = 0.06821$$

$$k_4 = hf(x_2 + h, u_2 + k_3) = (0.1) f(0.8, 1.34564) = (0.1) \cdot (0.6961) \\ = 0.06961$$

$$u_8 = u_2 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_8 = 1.27744 + \frac{1}{6} [0.06643 + 2(0.06822) + 2(0.06821) + (0.06961)]$$

$$u_8 = 1.3456$$

$$\underline{u(0.8) = 1.3456}$$

$$k_1 = hf(x_8, u_8) = (0.1)f(0.8, 1.3456) = (0.1) \cdot (0.69611) \\ = 0.06961$$

$$k_2 = hf\left(x_8 + \frac{h}{2}, u_8 + \frac{k_1}{2}\right) = (0.1)f(0.85, 1.3804) = (0.1) \cdot (0.70722) \\ = 0.07072$$

$$k_3 = hf\left(x_8 + \frac{h}{2}, u_8 + \frac{k_2}{2}\right) = (0.1)f(0.85, 1.38096) = (0.1) \cdot (0.70711) \\ = 0.07071$$

$$k_4 = hf(x_8 + h, u_8 + k_3) = (0.1)f(0.9, 1.41631) = (0.1) \cdot (0.71559) \\ = 0.07156$$

$$u_9 = u_8 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_9 = 1.3456 + \frac{1}{6} [0.06961 + 2(0.07072) + 2(0.07071) + (0.07156)]$$

$$u_9 = 1.41627$$

$$\underline{u(0.9) = 1.41627}$$

$$k_1 = hf(x_9, u_9) = (0.1) f(0.9, 1.41627) = (0.1) \cdot (0.71553) \\ = 0.07156$$

$$k_2 = hf\left(x_9 + \frac{h}{2}, u_9 + \frac{k_1}{2}\right) = (0.1) f(0.95, 1.45205) = (0.1) \cdot (0.72173) \\ = 0.07217$$

$$k_3 = hf\left(x_9 + \frac{h}{2}, u_9 + \frac{k_2}{2}\right) = (0.1) f(0.95, 1.45236) = (0.1) \cdot (0.72166) \\ = 0.07217$$

$$k_4 = hf(x_9 + h, u_9 + k_3) = (0.1) f(1, 1.48843) = (0.1) \cdot (0.72573) \\ = 0.07257$$

$$u_{10} = u_9 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$u_{10} = 1.41627 + \frac{1}{6} [0.07156 + 2(0.07217) + 2(0.07217) + (0.07257)]$$

$$u_{10} = 1.4884$$

$$u(1) = 1.4884$$

Heun Metodu

$$u_{n+1} = u_n + \frac{h}{2} (f(x_n, u_n) + f(x_{n+1}, \tilde{u}_{n+1}))$$

1. adım

$$x_1 = x_0 + h = 0 + \frac{1}{20} = \frac{1}{20}$$

$$\tilde{u} = u_0 + h f(x_0, u_0) = 1 + h f(0, 1) = (1) + \frac{1}{20} (0) = 1$$

$$\begin{aligned} u_1 &= u_0 + \frac{h}{2} (f(x_0, u_0) + f(x_1, \tilde{u}_1)) = 1 + \frac{h}{2} (f(0, 1) + f(\frac{1}{20}, 1)) \\ &= 1 + \frac{\frac{1}{20}}{2} (0 + (\frac{1}{20e} + \frac{20}{401})) \end{aligned}$$

$$= \underline{1.00170673209448}$$

2. adım

$$x_2 = x_1 + h = \frac{1}{20} + \frac{1}{20} = \frac{1}{10}$$

$$\begin{aligned} \tilde{u}_2 &= u_1 + h f(x_1, u_1) = 1.00170673209448 + \frac{1}{20} \cdot f(\frac{1}{20}, 1.00170673209448) \\ &= (1.00170673209448) + \frac{1}{20} (0.0682379169717956) \\ &= 1.00511862794307 \end{aligned}$$

$$\begin{aligned} u_2 &= u_1 + \frac{h}{2} (f(x_1, u_1) + f(x_2, \tilde{u}_2)) = 1.00170673209448 + \frac{h}{2} (f(\frac{1}{20}, 1.00170673209448) \\ &\quad + f(\frac{1}{10}, 1.00511862794307)) \\ &= 1.00170673209448 + \frac{\frac{1}{20}}{2} (0.0682379169717956 \\ &\quad + 0.13561002241584) = \boxed{1.00680299057917} \end{aligned}$$

3. dim

$$x_3 = x_2 + h = \frac{1}{10} + \frac{1}{20} = \frac{3}{20}$$

$$\begin{aligned} \tilde{u}_3 &= u_2 + h f(x_2, u_2) = 1.00680293057917 + h f\left(\frac{1}{10}, 1.00680293057917\right) = \\ &= (1.00680293057917) + \frac{1}{20} \cdot (0.135548428620698) = 1.01358035201021 \end{aligned}$$

$$\begin{aligned} u_3 &= u_2 + \frac{h}{2} (f(x_2, u_2) + f(x_3, \tilde{u}_3)) = 1.00680293057917 + \\ &+ \frac{h}{2} (f\left(\frac{1}{10}, 1.00680293057917\right) + f\left(\frac{3}{20}, 1.01358035201021\right)) \\ &= 1.00680293057917 + \frac{\frac{1}{20}}{2} \cdot (0.135548428620698 + 0.201136858365574) \\ &= \boxed{1.01522006275383} \end{aligned}$$

6. dim

$$x_4 = x_3 + h = \frac{3}{20} + \frac{1}{20} = \frac{1}{5}$$

$$\begin{aligned} \tilde{u}_4 &= u_3 + h f(x_3, u_3) = 1.01522006275383 + h f\left(\frac{3}{20}, 1.01522006275383\right) = \\ &= (1.01522006275383) + \frac{1}{20} \cdot (0.20104766960321) = 1.0252724423399 \end{aligned}$$

$$\begin{aligned} u_4 &= u_3 + \frac{h}{2} (f(x_3, u_3) + f(x_4, \tilde{u}_4)) = 1.01522006275383 + \\ &+ \frac{h}{2} (f\left(\frac{3}{20}, 1.01522006275383\right) + f\left(\frac{1}{5}, 1.0252724423399\right)) = \\ &= 1.01522006275383 + \frac{\frac{1}{20}}{2} \cdot (0.20104766960321 + 0.264047437502724) \\ &= \boxed{1.02684744043148} \end{aligned}$$

5. adim

$$x_5 = x_4 + h = \frac{1}{5} + \frac{1}{20} = \frac{1}{4}$$

$$\tilde{y}_5 = y_4 + h f(x_4, y_4) = 1.02684744043148 + h f\left(\frac{1}{5}, 1.02684744043148\right) =$$

$$= (1.02684744043148) + \frac{1}{20} (0.0263934536752664) = 1.04004416726911$$

$$y_5 = y_4 + \frac{h}{2} (f(x_4, y_4) + f(x_5, \tilde{y}_5)) = 1.02684744043148 + \frac{h}{2} (f\left(\frac{1}{5}, 1.02684744043148\right) +$$

$$+ f\left(\frac{1}{4}, 1.04004416726911\right)) = 1.02684744043148 + \frac{1}{20} (0.0263934536752664 +$$

$$+ (0.323657885440926)) = \boxed{1.04153715098632}$$

6. adim

$$x_6 = x_5 + h = \frac{1}{4} + \frac{1}{20} = \frac{3}{10}$$

$$\tilde{y}_6 = y_5 + h f(x_5, y_5) = 1.04153715098632 + h f\left(\frac{1}{4}, 1.04153715098632\right) =$$

$$= (1.04153715098632) + \frac{1}{20} (0.323522064174342) = 1.05771325419503$$

$$y_6 = y_5 + \frac{h}{2} (f(x_5, y_5) + f(x_6, \tilde{y}_6)) = 1.04153715098632 + \frac{h}{2} (f\left(\frac{1}{4}, 1.04153715098632\right) +$$

$$+ f\left(\frac{3}{10}, 1.05771325419503\right)) = 1.04153715098632 +$$

$$+ \frac{1}{20} (0.323522064174342 + (0.879404049769068))$$

$$= \boxed{1.0591103038349}$$

7. ad.m

$$x_7 = x_6 + h = \frac{3}{10} + \frac{1}{20} = \frac{7}{20}$$

$$\begin{aligned} \tilde{y}_7 &= u_6 + hf(x_6, u_6) = 1.0591103038349 + hf\left(\frac{7}{10}, 1.0591103038349\right) = \\ &= (1.0591103038349) + \frac{1}{20} \cdot (0.375258614167175) = 1.07807223454326 \end{aligned}$$

$$\begin{aligned} u_7 &= u_6 + \frac{h}{2} (f(x_6, u_6) + f(x_7, \tilde{y}_7)) = 1.0591103038349 + \\ &+ \frac{h}{2} (f\left(\frac{7}{10}, 1.0591103038349\right) + f\left(\frac{7}{20}, 1.07807223454326\right)) = 1.0591103038349 + \\ &+ \frac{1}{20} (0.375258614167175) + (0.430631675877526) = \boxed{1.07936406108852} \end{aligned}$$

8. ad.m

$$x_8 = x_7 + h = \frac{7}{20} + \frac{1}{20} = \frac{2}{5}$$

$$\begin{aligned} \tilde{y}_8 &= u_7 + hf(x_7, u_7) = 1.07936406108852 + hf\left(\frac{7}{20}, 1.07936406108852\right) = \\ &= (1.07936406108852) + \frac{1}{20} \cdot (0.636728053626886) = 1.10000036376986 \end{aligned}$$

$$\begin{aligned} u_8 &= u_7 + \frac{h}{2} (f(x_7, u_7) + f(x_8, \tilde{y}_8)) = 1.07936406108852 + \frac{h}{2} (f\left(\frac{7}{20}, 1.07936406108852\right) \\ &+ f\left(\frac{2}{5}, 1.00000036376986\right)) = 1.07936406108852 + \frac{1}{20} (0.636728053626886) + \\ &+ (0.477856111785993) = \boxed{1.10207891522409} \end{aligned}$$

9. adim

$$x_9 = x_8 + h = \frac{2}{5} + \frac{1}{20} = \frac{9}{20}$$

$$\begin{aligned} \tilde{u}_9 &= u_8 + h f(x_8, u_8) = 1.10207891522409 + h f\left(\frac{2}{5}, 1.10207891522409\right) = \\ &= (1.10207891522409) + \frac{1}{20} (0.477699502807764) = 1.12586383036948 \\ u_9 &= u_8 + \frac{h}{2} (f(x_8, u_8) + f(x_9, \tilde{u}_9)) = 1.10207891522409 + \frac{h}{2} (f\left(\frac{2}{5}, 1.10207891522409\right) + \\ &+ f\left(\frac{9}{20}, 1.12586383036948\right)) = 1.10207891522409 + \frac{\frac{1}{20}}{2} (0.477699502807764 + \\ &(0.52017323152475)) = \boxed{1.1270257336506} \end{aligned}$$

10. adim

$$x_{10} = x_9 + h = \frac{9}{20} + \frac{1}{20} = \frac{1}{2}$$

$$\begin{aligned} \tilde{u}_{10} &= u_9 + h f(x_9, u_9) = 1.1270257336506 + h f\left(\frac{9}{20}, 1.1270257336506\right) = \\ &(1.1270257336506) + \frac{1}{20} (0.520018337341410) = 1.15302665051767 \\ u_{10} &= u_9 + \frac{h}{2} (f(x_9, u_9) + f(x_{10}, \tilde{u}_{10})) = 1.1270257336506 + \frac{h}{2} (f\left(\frac{9}{20}, 1.1270257336506\right) + \\ &+ f\left(\frac{1}{2}, 1.15302665051767\right)) = 1.1270257336506 + \frac{\frac{1}{20}}{2} (0.520018337341410) + \\ &+ (0.55783994684269)) = \boxed{1.15397219045124} \end{aligned}$$

11. odim

$$x_{11} = x_{10} + h = \frac{1}{2} + \frac{1}{20} = \frac{11}{20}$$

$$\tilde{u}_{11} = u_{10} + h f(x_{10}, u_{10}) = 1.15397219045124 + h f\left(\frac{1}{2}, 1.15397219045124\right) =$$

$$= (1.15397219045124) + \frac{1}{20} \cdot (0.5576907611258762) = 1.18185672851418$$

$$\begin{aligned} u_{11} &= u_{10} + \frac{h}{2} (f(x_{10}, u_{10}) + f(x_{11}, \tilde{u}_{11})) = 1.15397219045124 + \frac{h}{2} (f\left(\frac{1}{2}, 1.15397219045124\right) + \\ &+ f\left(\frac{11}{20}, 1.18185672851418\right)) = 1.15397219045124 + \\ &+ \frac{\frac{1}{20}}{2} (0.5576907611258762 + (0.500854679342108)) = \boxed{1.18268832646639} \end{aligned}$$

12. odim

$$x_{12} = x_{11} + h = \frac{11}{20} + \frac{1}{20} = \frac{3}{5}$$

$$\tilde{u}_{12} = u_{11} + h f(x_{11}, u_{11}) = 1.18268832646639 + h f\left(\frac{11}{20}, 1.18268832646639\right) =$$

$$= (1.18268832646639) + \frac{1}{20} (0.590814455644) = 1.21222904924461$$

$$u_{12} = u_{11} + \frac{h}{2} (f(x_{11}, u_{11}) + f(x_{12}, \tilde{u}_{12})) = 1.18268832646639$$

$$+ \frac{h}{2} (f\left(\frac{11}{20}, 1.18268832646639\right) + f\left(\frac{3}{5}, 1.21222904924461\right)) =$$

$$= 1.18268832646639 + \frac{\frac{1}{20}}{2} (0.590814455644 + (0.619696464556128))$$

$$= \boxed{1.2129510994684}$$

13. adim

$$x_{13} = x_{12} + h = \frac{7}{5} + \frac{1}{20} = \frac{17}{20}$$

$$\begin{aligned} \tilde{y}_{13} &= y_{12} + h f(x_{12}, y_{12}) = 1.2129510944684 + h f\left(\frac{7}{5}, 1.2129510994634\right) = \\ &= (1.2129510994634) + \frac{1}{20} (0.619567610679437) = 1.2439294800337 \end{aligned}$$

$$\begin{aligned} y_{13} &= y_{12} + \frac{h}{2} (f(x_{12}, y_{12}) + f(x_{13}, \tilde{y}_{13})) = 1.2129510934634 + \frac{h}{2} (f\left(\frac{7}{5}, 1.2129510994684\right) \\ &+ f\left(\frac{17}{20}, 1.24392948000337\right)) = 1.2129510934684 + \frac{1}{20} (0.619567610679437) + \\ &+ (0.644704061708033)) = \boxed{1.24454783126909} \end{aligned}$$

14. adim

$$x_{14} = x_{13} + h = \frac{17}{20} + \frac{1}{20} = \frac{7}{10}$$

$$\begin{aligned} \tilde{y}_{14} &= y_{13} + h f(x_{13}, y_{13}) = 1.24454783126909 + h f\left(\frac{17}{20}, 1.24454783126909\right) = \\ &= (1.24454783126909) + \frac{1}{20} (0.644188230320012) = 1.27675730278509 \end{aligned}$$

$$\begin{aligned} y_{14} &= y_{13} + \frac{h}{2} (f(x_{13}, y_{13}) + f(x_{14}, \tilde{y}_{14})) = 1.24454783126909 \\ &+ \frac{h}{2} (f\left(\frac{17}{20}, 1.24454783126909\right) + f\left(\frac{7}{10}, 1.27675730278509\right)) = \\ &= 1.24454783126909 + \frac{1}{20} (0.644188230320012 + (0.665056905943233)) \end{aligned}$$

$$\boxed{= 1.27727901067567}$$

15. adim

$$x_{15} = x_{14} + h = \frac{2}{10} + \frac{1}{20} = \frac{3}{4}$$

$$\begin{aligned} \tilde{u}_{15} &= u_{14} + h f(x_{14}, u_{14}) = 1.27727901967567 + h f\left(\frac{2}{10}, 1.27727901967567\right) = \\ &= (1.27727901967567) + \frac{1}{20} \cdot (0.664955062986014) = 1.31052677282407 \end{aligned}$$

$$u_{15} = u_{14} + \frac{h}{2} (f(x_{14}, u_{14}) + f(x_{15}, \tilde{u}_{15})) = 1.27727901967567$$

$$+ \frac{h}{2} (f\left(\frac{2}{10}, 1.27727901967567\right) + f\left(\frac{3}{4}, 1.31052677282407\right)) = 1.27727901967567 +$$

$$\frac{1}{20} (0.664955062986014 + 0.682258460355693) = \boxed{1.31095935799921}$$

16. adim

$$x_{16} = x_{15} + h = \frac{3}{4} + \frac{1}{20} = \frac{4}{5}$$

$$\tilde{u}_{16} = u_{15} + h f(x_{15}, u_{15}) = 1.31095935799921 + h f\left(\frac{3}{4}, 1.31095935799921\right) =$$

$$= (1.31095935799921) + \frac{1}{20} \cdot (0.682170994861793) = 1.3450679077423$$

$$u_{16} = u_{15} + \frac{h}{2} (f(x_{15}, u_{15}) + f(x_{16}, \tilde{u}_{16})) = 1.31095935799921 +$$

$$+ \frac{h}{2} (f\left(\frac{3}{4}, 1.31095935799921\right) + f\left(\frac{4}{5}, 1.3450679077423\right)) =$$

$$1.31095935799921 + \frac{1}{20} (0.682170994861793 + 0.696222490686665) =$$

$$\boxed{1.34541919513792}$$

17. adım

$$x_{17} = x_{16} + h = \frac{4}{5} + \frac{1}{20} = \frac{17}{20}$$

$$\begin{aligned} u_{17} &= u_{16} + h f(x_{16}, u_{16}) = 1.34541919513792 + h f\left(\frac{4}{5}, 1.34541919513792\right) \\ &= (1.34541919513792) + \frac{1}{20} (0.696149283064476) = 1.38022665959115 \end{aligned}$$

$$\begin{aligned} u_{17} &= u_{16} + \frac{h}{2} (f(x_{16}, u_{16}) + f(x_{17}, \tilde{u}_{17})) = 1.34541919513792 + \\ &+ \frac{h}{2} (f\left(\frac{4}{5}, 1.34541919513792\right) + f\left(\frac{17}{20}, 1.38022665959115\right)) = \\ &= 1.34541919513792 + \frac{1}{20} (0.696149283064476 + (0.707262101669793)) \end{aligned}$$

$$\boxed{= 1.38050447930605}$$

18. adım

$$x_{18} = x_{17} + h = \frac{17}{20} + \frac{1}{20} = \frac{9}{10}$$

$$\begin{aligned} u_{18} &= u_{17} + h f(x_{17}, u_{17}) = 1.38050447930605 + h f\left(\frac{17}{20}, 1.38050447930605\right) = \\ &= (1.38050447930605) + \frac{1}{20} (0.707202713787077) = 1.41586461559541 \end{aligned}$$

$$\begin{aligned} u_{18} &= u_{17} + \frac{h}{2} (f(x_{17}, u_{17}) + f(x_{18}, \tilde{u}_{18})) = 1.38050447930605 + \\ &+ \frac{h}{2} (f\left(\frac{17}{20}, 1.38050447930605\right) + f\left(\frac{9}{10}, 1.41586461559541\right)) = \\ &= 1.38050447930605 + \frac{1}{20} (0.707202713787077 + (0.71568166931525)) \end{aligned}$$

$$\boxed{= 1.41607658948361}$$

(34)

19. adim

$$x_{19} = x_{18} + h = \frac{9}{20} + \frac{1}{20} = \frac{19}{20}$$

$$\begin{aligned} \tilde{u}_{19} &= u_{18} + h f(x_{18}, u_{18}) = 1.41607658948361 + h f\left(\frac{9}{20}, 1.41607658948361\right) = \\ &= (1.41607658948361) + \frac{1}{20} \cdot (0.71563536977284) = 1.45185835797248 \end{aligned}$$

$$\begin{aligned} u_{19} &= u_{18} + \frac{h}{2} (f(x_{18}, u_{18}) + f(x_{19}, \tilde{u}_{19})) = 1.41607658948361 + \\ &+ \frac{h}{2} (f\left(\frac{9}{20}, 1.41607658948361\right) + f\left(\frac{19}{20}, 1.45185835797248\right)) = \\ &= 1.41607658948361 + \frac{1}{20} (0.71563536977284 + (0.721971008232136)) = \end{aligned}$$

$$= 1.45201174893385$$

20. adim

$$x_{20} = x_{19} + h = \frac{19}{20} + \frac{1}{20} = 1$$

$$\begin{aligned} \tilde{u}_{20} &= u_{19} + h f(x_{19}, u_{19}) = 1.45201174893385 + h f\left(\frac{19}{20}, 1.45201174893385\right) = \\ &= (1.45201174893385) + \frac{1}{20} (0.721936892338078) = 1.4880021094925 \end{aligned}$$

$$\begin{aligned} u_{20} &= u_{19} + \frac{h}{2} (f(x_{19}, u_{19}) + f(x_{20}, \tilde{u}_{20})) = 1.45201174893385 + \\ &+ \frac{h}{2} (f\left(\frac{19}{20}, 1.45201174893385\right) + f(1, 1.4880021094925)) = 1.45201174893385 + \\ &+ \frac{1}{20} (0.721936892338078 + (0.725801588217341)) = 1.48820021094925 \end{aligned}$$

Sonuçların Kıyaslanması

$$u(1) = 1.48820021094925$$

	Euler	Heun	Runge-Kutta
h	0.025	0.05	0.1
u(1)	1.48024	1.48820021094925	1.4884

(35)

Soru 5)

1. yckleim

$$x_1 = \frac{1}{4} [0 + (0) - 2(0)] = \frac{1}{4} [0] = 0$$

$$y_1 = \frac{1}{6} [1 + (0) + (0)] = \frac{1}{6} [1] = 0,16667$$

$$z_1 = \frac{1}{5} [-8 - 2(0) + (0,16667)] = \frac{1}{5} [-7,83333] = -1,56667$$

2. yckleim

$$x_2 = \frac{1}{4} [0 + (0,16667) - 2(-1,56667)] = \frac{1}{4} [3,3] = 0,825$$

$$y_2 = \frac{1}{6} [1 + (0,825) + (-1,56667)] = \frac{1}{6} [0,25833] = 0,04306$$

$$z_2 = \frac{1}{5} [-8 - 2(0,825) + (0,04306)] = \frac{1}{5} [-9,60964] = -1,92139$$

3. yckleim

$$x_3 = \frac{1}{4} [0 + (0,04306) - 2(-1,92139)] = \frac{1}{4} [3,88583] = 0,97146$$

$$y_3 = \frac{1}{6} [1 + (0,97146) + (-1,92139)] = \frac{1}{6} [0,05007] = 0,00834$$

$$z_3 = \frac{1}{5} [-8 - 2(0,97146) + (0,00834)] = \frac{1}{5} [-9,94558] = -1,98912$$

Devamı
//

4. yeklesim

$$x_4 = \frac{1}{4} [0 + (0,00834) - 2(-1,98691)] = \frac{1}{4} [3,98217] = 0,99554$$

$$y_4 = \frac{1}{6} [1 + (0,99554) + (-1,98691)] = \frac{1}{6} [0,00863] = 0,00144$$

$$z_4 = \frac{1}{5} [-8 - 2(0,99554) + (0,00144)] = \frac{1}{5} [-9,98965] = -1,99793$$

5. yeklesim

$$x_5 = \frac{1}{4} [0 + (0,00144) - 2(-1,99793)] = \frac{1}{4} [3,99732] = 0,99932$$

$$y_5 = \frac{1}{6} [1 + (0,99932) + (-1,99793)] = \frac{1}{6} [0,00139] = 0,00023$$

$$z_5 = \frac{1}{5} [-8 - 2(0,99932) + (0,00023)] = \frac{1}{5} [-9,99842] = -1,99968$$

6. yeklesim

$$x_6 = \frac{1}{4} [0 + (0,00023) - 2(-1,99968)] = \frac{1}{4} [3,9996] = 0,9999$$

$$y_6 = \frac{1}{6} [1 + (0,9999) + (-1,99968)] = \frac{1}{6} [0,00022] = 0,00004$$

$$z_6 = \frac{1}{5} [-8 - 2(0,9999) + (0,00004)] = \frac{1}{5} [-9,99976] = -1,99995$$

$$x_1 = 0,9999 \cong 1$$

$$x_2 = 0,00004 \cong 0$$

$$x_3 = -1,99995 \cong -2$$

$$h_{x_1} = |x_6 - x_5| = 0,00058 < 0,001$$

$$= |y_6 - y_5| = 0,00013 < 0,001$$

$$= |z_6 - z_5| = 0,00027 < 0,001$$